

The empirical recurrence for OEIS A260839, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A260839 records “Empirical recurrence of order 92” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 146$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 97$. Its last coefficient is $c_{92} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A260839 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 00000011 or 00010101.”. It has offset 1 and begins

1320, 1645, 3470, 6232, 8646, 19718, 37882, 58489, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:11:58, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 146$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 146$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 292$ exact terms from $n = 5$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	0	c_{24}	98795	c_{47}	49604	c_{70}	1082
c_2	1	c_{25}	160957	c_{48}	−164254	c_{71}	−53789
c_3	12	c_{26}	103421	c_{49}	−241760	c_{72}	−3827
c_4	19	c_{27}	−50063	c_{50}	−110857	c_{73}	−1459
c_5	−11	c_{28}	−213567	c_{51}	23789	c_{74}	−4458
c_6	−78	c_{29}	−256926	c_{52}	269188	c_{75}	18898
c_7	−197	c_{30}	−99241	c_{53}	289141	c_{76}	−1694
c_8	−119	c_{31}	139619	c_{54}	86155	c_{77}	−262
c_9	341	c_{32}	303817	c_{55}	−126023	c_{78}	2310
c_{10}	1030	c_{33}	272976	c_{56}	−280119	c_{79}	−4747
c_{11}	1392	c_{34}	38829	c_{57}	−211398	c_{80}	1005
c_{12}	−62	c_{35}	−199553	c_{58}	−20711	c_{81}	322
c_{13}	−3520	c_{36}	−276534	c_{59}	186163	c_{82}	−579
c_{14}	−6592	c_{37}	−177431	c_{60}	196250	c_{83}	775
c_{15}	−5335	c_{38}	20406	c_{61}	99567	c_{84}	−230
c_{16}	4675	c_{39}	172315	c_{62}	−14658	c_{85}	−84
c_{17}	19523	c_{40}	136476	c_{63}	−170346	c_{86}	75
c_{18}	25577	c_{41}	33684	c_{64}	−96470	c_{87}	−70
c_{19}	10407	c_{42}	−43226	c_{65}	−33194	c_{88}	22
c_{20}	−29130	c_{43}	−103537	c_{66}	10976	c_{89}	8
c_{21}	−68493	c_{44}	24472	c_{67}	111715	c_{90}	−3
c_{22}	−64052	c_{45}	109520	c_{68}	31289	c_{91}	2
c_{23}	−1031	c_{46}	74593	c_{69}	8499	c_{92}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 97$, and it is the recurrence OEIS A260839 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1x^{92-1} - \dots - c_{92}$. Its constant term is $-c_{92} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A260839.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.